

Janus 5.0 — Master Mathematical & Structural Standard

[Your Name]

Version 0.1 — July 2025

Contents

1	Introduction & Methodological Notes	3
1.1	Objective	3
1.2	Scope	3
1.3	On AI-Augmented Research	3
1.4	Notation	3
2	Core Variables & Data Schema	3
2.1	Node Structure	3
2.2	Edge Structure	3
2.3	Global Graph Variables	3
3	Derived Metrics & Formulas	4
3.1	Contradiction Density	4
3.2	Load Pressure	4
3.3	Coherence Mass	4
3.4	Projection Bias	4
3.5	Graph Entropy	4
4	Core Functions & Algorithms	4
5	Parameters & Thresholds	5
6	Data Structure & JSON Standards	5
7	Support Utilities	5
8	Future Expansion Modules	5
9	Glossary of Terms	5
10	Version Notes	6

Contents

1 Introduction & Methodological Notes

1.1 Objective

This document formalizes the mathematical architecture, data schemas, and dynamic algorithms underpinning the Janus 5.0 framework — a cognitive recursion engine designed to model psychological structures as evolving directed graphs.

1.2 Scope

Defines atomic data types (nodes, edges), global aggregates, derived metrics (entropy, contradiction density, coherence mass, projection bias), transformation functions, safety parameters, and serialization standards. Excludes UI, hardware interfaces, or neurobiological layers.

1.3 On AI-Augmented Research

AI systems were used to accelerate drafting and multi-branch analytical expansions, serving strictly as amplifiers of exploration. All theoretical constructs and mathematical formulations originate from first-principles inquiry, subsequently extended through computational assistance.

1.4 Notation

Scalars: w, p ; Vectors: $\mathbf{p}$; Graphs: G, S, C ; Functions: $\text{function}(\cdot)$. Entropy uses natural logs unless noted.

2 Core Variables & Data Schema

2.1 Node Structure

$$n = (\text{id}, \text{type}, w_b, s, d, p, L)$$

where each field is precisely typed and interpreted.

2.2 Edge Structure

$$e = (\text{source}, \text{target}, w, t)$$

2.3 Global Graph Variables

$\bar{s}$: global stability; $\bar{c}$: average contradiction density; β : projection bias; $\bar{d}$: average recursion depth; $H(G)$: entropy.

3 Derived Metrics & Formulas

3.1 Contradiction Density

$$\text{contradiction_density}(C) = \frac{\sum_{e \in E_C, t_e = \text{contradict}} w_e}{\sum_{e \in E_C} w_e}$$

with default to 0 if denominator is zero.

3.2 Load Pressure

$$\text{load_pressure}(n) = \sum_{e \in \text{in}(n), t_e \in \{\text{contradict}, \text{emotion}\}} |w_e|$$

3.3 Coherence Mass

$$\text{coherence_mass}(S) = \frac{1}{H(S) + \epsilon}$$

3.4 Projection Bias

$$\text{projection_bias}(G) = \frac{\sum_{e \in E, t_e = \text{projects_to}} w_e}{\sum_{e \in E} w_e}$$

3.5 Graph Entropy

$$\text{graph_entropy}(S) = - \sum_i p_i \log p_i, \quad p_i = \frac{w_i}{\sum_j w_j}$$

4 Core Functions & Algorithms

Formal signatures and brief definitions for:

- `calculate_contradiction_density(C)`
- `compute_load_pressure(n)`
- `calculate_coherence_mass(S)`
- `calculate_projection_bias(G)`
- `graph_entropy(S)`

Also includes transformations like `apply_contradiction_injection`, normalization, recursive checks, flag instability, snapshotting, rollback, audit logs.

5 Parameters & Thresholds

Key parameters with typical defaults:

Parameter	Symbol	Default	Description
Max recursion depth	$d_{\max}$	10	Caps introspection depth.
Paradox tolerance	θ_c	0.35	Max safe average contradiction.
Load pressure threshold	θ_p	0.65	Max local strain before interventions.
Entropy collapse cutoff	θ_m	0.25	Min coherence mass.
Projection bias alert	θ_β	0.8	Flags excessive forward projection.
Rollback on flag	r_f	True	Auto-rollback control.
Epsilon stabilizer	ϵ	0.001	Protects inverse entropy division.

6 Data Structure & JSON Standards

Canonical JSON structures for nodes, edges, global snapshots, and full system states.

7 Support Utilities

Includes `snapshot_state`, `rollback_to_snapshot`, `audit_change_log`, `validate_graph_integrity`, `compare_snapshots`.

8 Future Expansion Modules

Proposals for:

- Multi-path scenario forecasts
- Core schema extraction
- Ambiguity saturation tests
- Entropy gradient mapping
- Simulated recovery protocols
- Meta-experiment audits

9 Glossary of Terms

Rigorous definitions of $w_b, s, d, p, H(S), \bar{s}, \bar{c}, \beta, \epsilon$, function roles and edge types.

10 Version Notes

Version	Date	Summary
0.1	July 2025	Complete initial mathematical and operational standard.